Weekly Meeting

25.08.15

Geon Lee

Alpha-value tuning

- Generate w. trained MLP

$$\ell_i = f_{\mathrm{LLM}}(x_{1:t-1})[i], \quad i = 1, \ldots, V$$

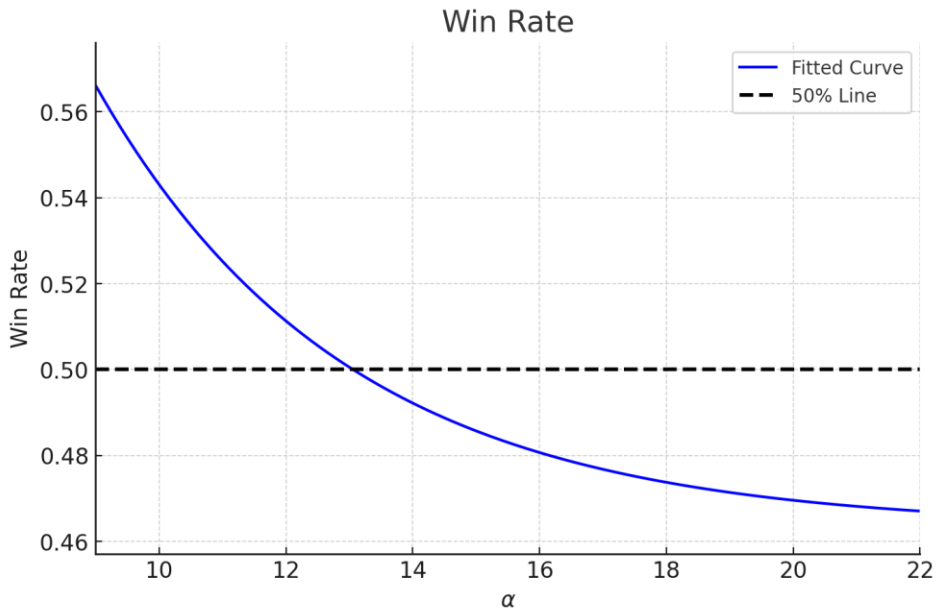
$$\ell'_i = \begin{cases} \ell_i + \alpha\, s_i, & i \in \mathcal{K} \\ \ell_i, & i \notin \mathcal{K} \end{cases}$$

  - s_i: MLP output from last hidden layer & next-token embedding

- Using the original model
  - 8192 → 8 → 1
  - Dataset: alpaca eval

Alpha-value tuning

- Alpha value:
  - $\alpha \approx 1$: No significant difference from baseline
  - $\alpha \geq 10$: Significant change
  - $\alpha \gg 10$: Large α → lower win rates



Dataset Expansion

- Chatbot Arena
  - lmsys/chatbot_arena_conversations
  - Randomly sample 2000 queries
    - (only English, one-turn conversation)

- Other candidates
  - Consider OpenHermes or others
  - Dataset contains a lot of noise
  - Not similar to AlpacaEval dataset

Dataset Expansion

- Chatbot arena:
  - 3498 pair-wise dataset (Exclude cycles. i.e. a>b>c>a)
  - 3298 train / 200 valid / 283 test

- Total (Alpaca eval + Chatbot arena):
  - 4914 pair-wise dataset (2457 instruction)
  - 4156 train / 484 valid / 484 test

Dataset Expansion

- Pair-wise test:
  - Generate: llama3.1, gpt-4o-mini (same setting as before)
  - Llama vs gpt win rate:
    - 587:1412
    - 572:1427
    - Detect preference bias (Include quality bias)
  - Llama vs Llama:
    - 1021:978

MLP train - 1

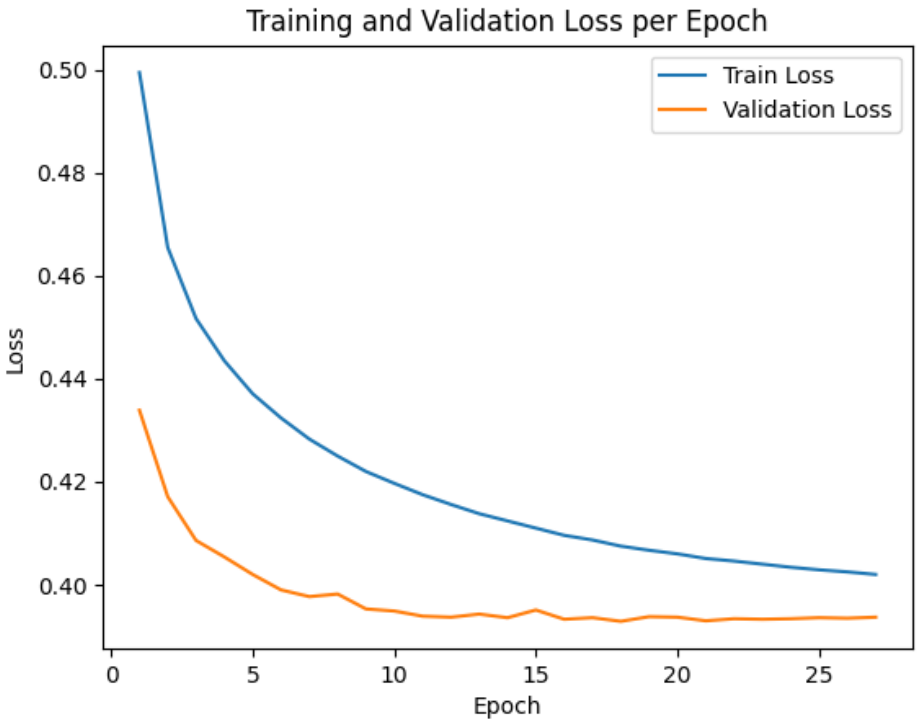
- Hyper-Parameters
  - Optimizer: SGD
  - Momentum: 0.9
  - Learning rate: 1×10^{-4}
  - Weight decay: 1×10^{-4}
  - Epochs: 50
  - Dropout: 0.25
  - Hidden width: 8

- Input dimension: 8192
  - (8192, 8) → GELU → (8, 1)

MLP train - 1

- Results
  - Improvement in accuracy
  - No dramatic performance improvement

```
ep27  train 0.4020 | val 0.3937  acc 64.703%
■ Early stopping triggered
[TEST] loss 0.3885 | acc 63.876%
```



MLP train - 2

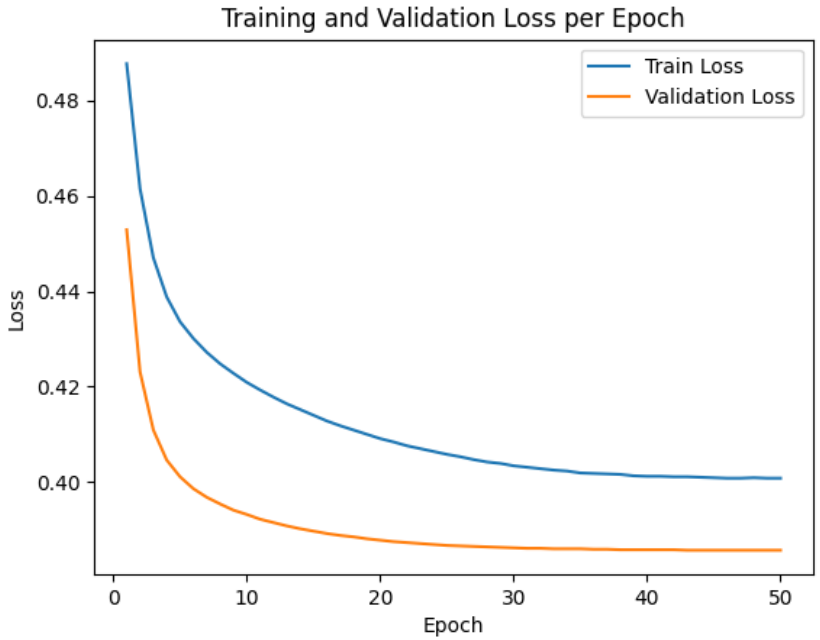
- Hyper-Parameters
  - Optimizer: SGD
  - Momentum: 0.9
  - Learning rate: 1×10^{-4}
  - Weight decay: 1×10^{-4}
  - Epochs: 50
  - Dropout: 0.2
  - Hidden width: 256 - 128

- Input dimension: 8192
  - (8192, 256) → GELU → (256, 128) → GELU → (128, 1)

MLP train - 2

- Results
  - Improvement in accuracy
  - No dramatic performance improvement
  - Parameter ↑ → No change in accuracy

```
ep50  train 0.4008 | val 0.3857  acc 64.926%
  ↳ saved pref_head_reg.pt
[TEST] loss 0.3969 | acc 63.612%
```



Discussion & TODO

- Hyper parameter tuning
  - In progress
  - Increase layers & width
- Num of parameters ↔ accuracy
  - Why performance is similar?
- Num of train datasets ↔ accuracy
  - Training dataset: nearly 4× larger
  - Why performance increases only slightly?

Weekly Meeting

25.08.22

Geon Lee

Experiment setting

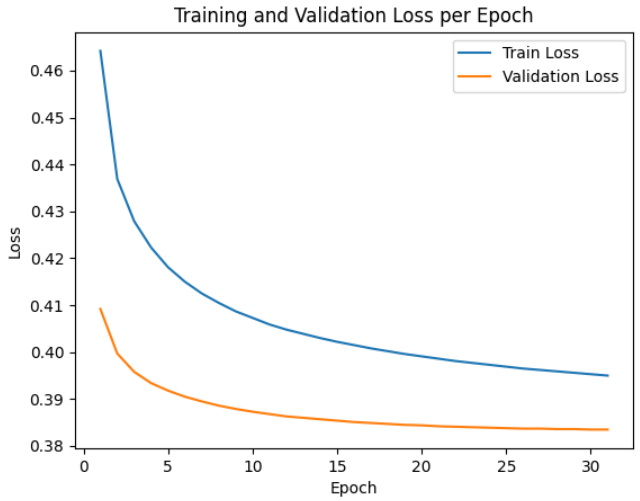
- Train set
  - 3956 pair
  - 3,073,005 tokens
- Valid set
  - 484 pair
  - 392,835 tokens
- Test set
  - 484 pair
  - 382,193 tokens

Increase parameter

- 3layer
  - 8192 → 1024 → 512 → 256 → 1
  - 8192 → 512 → 256 → 128 → 1
  - etc
- Skip connection
  - 8192 → last layer
- Optimizer
  - AdamW
  - SGD

Increase parameter

- Results
  - Still no dramatic changes
  - Accuracy 69%: maybe outlier

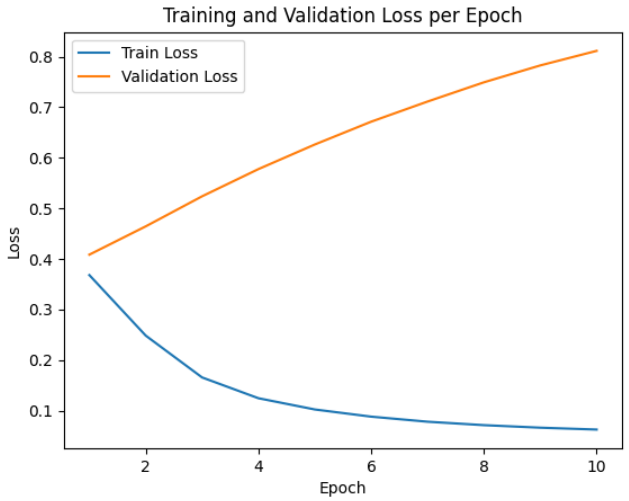


*512 → 256 → 128

```
ep31  train 0.3851 | val 0.3866  acc 64.657%
■ Early stopping triggered
Loading best model from pref_head_reg_layer.pt for final testing...
[TEST] loss 0.3901 | acc 64.132%
```

*last saved

```
ep22  train 0.3905 | val 0.3864  acc 64.698%
  ↳ saved pref_head_reg_layer.pt
```



*1024 → 512 → 256

```
ep10  train 0.0562 | val 0.8251  acc 60.963%
■ Early stopping triggered
Loading best model from pref_head_reg_layer_512 for final testing...
[TEST] loss 0.3430 | acc 69.218%
```

*last saved

```
ep01  train 0.3682 | val 0.4086  acc 63.523%
  ↳ saved pref_head_reg_layer_512
```

Another hidden layer

- Llama3.1-8B-inst
  - 32 attention layer

- Input dimension:
  - 30th layer (4096)
  - 24th layer (4096)
  - Next token embedding (4096)

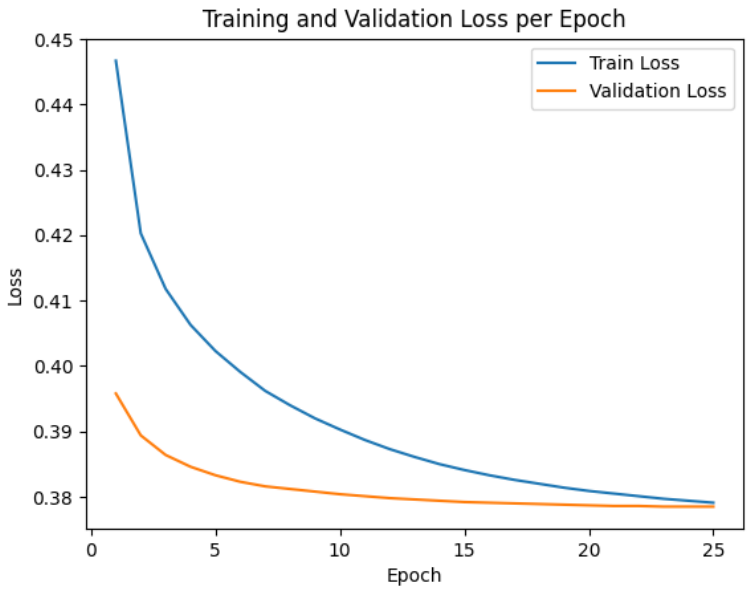
Another hidden layer

- Architecture
  - 3 layer
  - SGD, AdamW
  - Width:
    - 512 → 256 → 128
    - 256 → 128 → 64
    - 64 → 32 → 16
    - 32 → 16 → 8
  - Skip connection:
    - 12288 → last layer
  - Input dimension: 12288

Another hidden layer

- Results
  - Architecture
    - 12288 → 64 → 32 → 16 → 1
    - 12288 → 16 → 1
  - Optimizer
    - SGD

- Still no changes in test set

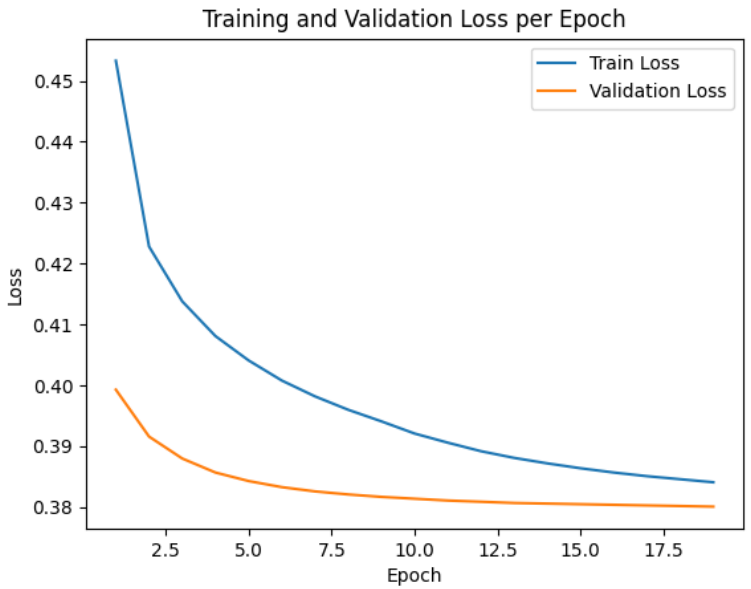



```
ep25  train 0.3765 | val 0.3807  acc 65.574%
■ Early stopping triggered
Loading best model from 3l_64w_12288d_sgd.pt for final testing...
[TEST] loss 0.3981 | acc 63.603%
```

```
ep16  train 0.3828 | val 0.3800  acc 65.593%
  ↳ saved 3l_64w_12288d_sgd.pt
```

Another hidden layer

- Results
  - Architecture
    - 12288 → 32 → 16 → 8 → 1
    - 12288 → 8 → 1
  - Optimizer
    - SGD




```
[TEST] loss 0.3995 | acc 63.488%
```

Discussion

- Same token → low accuracy
- Generalize
- Increase input dimension → dataset size / parameter
  - Params: 789,121
  - Train tokens: 3,073,005 (but 7,912 win-lose set)
- Another hidden layer